Dissertation

Demand Forecasting And Inventory Management

Course No.: S1-26_SSZG628T

Course Title: Dissertation

Dissertation Title: Demand Forecasting In Inventory Management

Dissertation Done by:

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- **BITS ID:** 2024MT13052
- **Degree Program:** M.Tech In Software Systems
- **Research Area:** Predictive Analysis

Dissertation carried out at:

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Broad Area of Work

The primary research area for this dissertation is Predictive Analytics and Machine Learning. The objective is to transition aircraft spare parts management from reactive component replenishment to a predictive maintenance inventory model. By leveraging historical flight service logs, aircraft configuration data, and environmental operational profiles (e.g., cold/snow landing environments versus temperate routes), this project builds an end-to-end analytical pipeline that forecasts component degradation, predicts required replacement parts, and formulates dynamic reordering parameters to share directly with suppliers via an interactive dashboard.

2. Background

In aerospace service departments, maintaining fleet readiness requires ensuring spare parts are prepositioned at repair stations prior to scheduled or unscheduled maintenance events. Under-procuring critical aviation components causes Aircraft On Ground (AOG) events, incurring severe financial penalties and operational disruptions. Conversely, over-procuring high-value flight components ties up millions in capital holding costs. Traditional parts demand forecasting relies on static flight-hour thresholds that fail to account for operational variances.

For instance, two aircraft sharing identical structural configurations experience vastly different component wear if one operates predominantly in harsh, freezing conditions compared to one operating in mild climates. To optimize supplier lead times and prevent AOG delays, we require a predictive framework that analyzes operational telemetry, flight cycles, environmental severity indices, and maintenance intervals to provide suppliers with accurate, look-ahead parts demand forecasts.

3. Objectives

Implement a dual-matrix ABC/XYZ Analysis to categorize aircraft spare parts by financial criticality (ABC) and demand predictability, prioritizing computational resources on high-value, variable-demand components.

Train and evaluate a Machine Learning model to predict required spare parts quantities, incorporating features such as total flight cycles, fleet age, maintenance schedules, and environmental operational flags.

Apply Market Basket Analysis using algorithms on historical maintenance teardown logs to discover companion part co-replacement rules (e.g., ensuring associated seal kits and structural bolts are bundled whenever a landing gear actuator is flagged).

4. Scope of Work

This project will utilize open-source aviation maintenance and flight operation datasets (commercial fleet operational logs, or simulated landing gear predictive maintenance datasets). Data preprocessing will involve grouping maintenance logs by Number, Configuration, and Flight Hours. Model execution will involve setting statistical baseline error rates, executing Apriori rule mining to map companion parts, and training a LightGBM regressor on Category A and B parts. The outputs will feed an interactive supplier-facing dashboard displaying predictive parts demand curves, ROP thresholds, and recommended batch order sizes. To keep the project strictly bounded and achievable the following elements are explicitly excluded: Live streaming of real-time flight telemetry or live weather APIs to avoid architectural latency and third-party API deprecations. This will also not include multiple parts, the scope will be limited to one part.

5. Plan of Work

Phases	Start Date-End Date	Work to be done
Dissertation Outline	26-July-2026 - 07-Aug-2026	Finalize project scope and prepare dissertation outline
Exploratory Data Analysis & Feature Engineering	03-Aug-2026 – 23-Aug-2026	Dataset observation, cleaning, extracting flight features, calculating Environmental factors and

Phases	Start Date-End Date	Work to be done
		completing inventory matrix classification
Baseline Formulations & Association Mining	24-Aug-2026 – 18-Sep-2026	Construct point moving average baselines, record benchmark, and run the Apriori algorithm to extract companion maintenance kit rules.
Machine Learning & Optimization Engine	19-Sep-2026 – 11-Oct-2026	Train and tune the model on aircraft parts, evaluate forecast accuracy, and program mathematical equations.
Dashboard and Frontend development	12-Oct-2026 – 20-Oct-2026	Construct the interactive supplier-facing dashboard interface.
Dissertation Review	20-Oct-2026 – 31-Oct-2026	Draft the full dissertation report
Final Submission	01-Nov-2026 – 08-Nov-2026	Finalize formatting and submit for review.

6. Literature References

1. Agrawal, R., & Srikant, R. (1994). Fast algorithms for mining association rules. Proceedings of the 20th International Conference on Very Large Data Bases (VLDB), 487–499.
2. Bacchetti, A., & Saccani, N. (2012). Spare parts classification and demand forecasting for stock control: Investigating the gap between research and practice. Omega, 40(6),

722–737.

3. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. Advances in Neural Information Processing Systems, 30, 3149–3157.
4. Lagoda, L., & Klumpp, M. (2024). Efficient warehouse and inventory management: The modified ABC-XYZ analysis as a framework to integrate demand forecasting and inventory control. In Dynamics in Logistics (LDIC 2024), Lecture Notes in Logistics. Springer.

7. Particulars of the Supervisor and Examiner

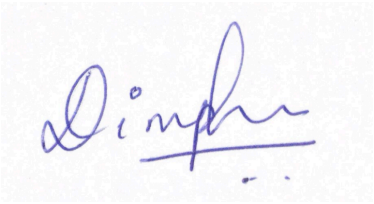
Field	Supervisor	Additional Examiner
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8. Remarks of the Supervisor

The proposed dissertation topic – predictive spare-parts demand forecasting for aircraft maintenance is directly relevant to supply chain and aviation. The scope, combining classification, model-based demand forecasting, and companion-part association mining, is well-defined and achievable within the semester timeline. The plan of work is approved, and I will provide periodic guidance during each phase.

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION
SECOND SEMESTER OF ACADEMIC YEAR 2026-2027
S1-26_SSZG628T : Dissertation OUTLINE

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